GG-C-1433B
July 12, 1976
SUPERSEDING
Int. Fed.Spec. GG-C-001433a(DSA-DM)
April 10, 1969

FEDERAL SPECIFICATION

CHISELS (CUTTING INSTRUMENTS), DENTAL

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers dental chisels used in various steps of cavity preparation.

1.2 Classification. Chisels shall be of the following types and numbers, as specified (see 6.1):

Type I. - Black's, Single Ended

No. 81
No. 83
No. 84
No. 85
No. 86

Type II. - Wedelstaedt's, Single Ended

No. 41
No. 42

Type III. - Wedelstaedt's (modified), Double Ended

No. 3 and 4
No. 5 and 6

Type IV. - Chisel, Double Ended

No. 40 and 41

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 6520

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ALTERNATE DESIGN

Federal Specifications:

GG-I-526	-	Instruments, Dental and Surgical; General Specifications.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup

Federal Standard:

Fed. Std. No. 123 - Marking For Shipment (Civil agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

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Drawings:

Defense Personnel Support Center (DPSC)

2208 - Chisels, Dental, Black.
20410 - Chisel, Dental, Wedelstaedt.
24177 - Chisel, Dental.

(Copies of Military Specifications, Standards, and Drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society For Testing and Materials Standards: (ASTM)

E 18 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

E 92 - Standard Method of Test for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 General requirements. The chisels shall be in accordance with the applicable requirements of GG-I-526 with the options and additional requirements specified herein.

3.2 Standard sample. The finish and workmanship of the blade shall be equal to or better than the standard sample (see 3.6.1, 3.8 and 6.2.)

3.3 Material. Chisels shall be fabricated of class 4, martensitic stainless steel, type 440A.

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request, at any time, or from time to time, during the

3.4 Hardness. Operative ends shall have a Rockwell hardness of C52 to C56 after proper stress relieving. Handles and shanks shall be properly tempered and shall have a hardness suitable for the purpose intended. Hardness shall be determined as specified in 4.3.1.

3.5 Style, design, and dimensions. Chisels shall be of one piece construction. Type I and type II chisels shall be single ended chisels. Type III and type IV chisels shall be double ended chisels. Type I chisels shall be of Dr. Black's design. Type II chisels shall be of Dr. Wedelstaedt's design. Type III chisels shall be of Dr. Wedelstaedt's design modified by Dr. Ferrier. Type IV chisel shall be as modified by Dr. Ferrier. The style, design, and dimensions of the chisels shall be in accordance with Defense Personnel Support Center (DPSC) drawings listed in table I for the applicable chisel type and number. The chisels shall conform to the applicable Black's formula as specified in the applicable drawing and as specified in table II herein. Chisels shall be suitable for cutting, planing, and cleaving enamel and dentin during dental operative procedures. Chisel handles shall be straight.

TABLE I. Drawing Numbers

Type	Chisel Number	DPSC Drawing
I	No. 81	2208
I	No. 83	2208
I	No. 84	2208
I	No. 85	2208
I	No. 86	2208
II	No. 41	20410
II	No. 42	20410
III	No. 3 and 4	24177
III	No. 5 and 6	24177
IV	No. 40 and 41	24177

3.5.1 Drawings. The drawings forming a part of this specification are not production drawings. Drawings establish dimensional limitations and control design and configuration. The supplier shall be responsible for preparing his own shop drawings. The supplier shall select dimensions and tolerances within the limitations prescribed on the drawings to provide a product with proper design, configuration and function. No deviations from the dimensional limitations prescribed on the drawings is permissible without prior approval of the contracting officer.

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3.6 Finish

3.6.1 Operative ends. The operative ends (blades) of the chisels, including sharpened edges of the beveled portion, shall have a fine honed finish such as that produced by using grade FF emery on a leather wheel. The chisel blades shall have sharp cutting edges and shall be free of defects such as coarse grind marks, cracks, burrs, deep pits, or feathers when viewed as specified in 4.2.1. The finish of the blade shall be equal to or better than the standard sample (see 3.2 and 6.2).

3.6.2 Handles and shanks. Handles and shanks of chisels shall have a mirror or bright polished finish. Handles shall have 40 ± 10 fillet or notch-type serrations per inch on seven faces and one smooth face as shown in the applicable DPSC drawing as specified in table I. Depth of serrations shall be 0.002 to 0.004 inch.

3.7 Identification marking. The chisels shall be permanently and legibly marked on the smooth face of the handle. In addition to the requirements of GG-I-526, each chisel shall be permanently and legibly marked with the respective instrument number and Black's formula, if applicable, as specified in table II. The respective instrument number for single ended chisels, types I and II, shall be placed on the handle no less than 5 millimeters (mm) nor more than 12 mm from the proximal end (free end). The respective instrument number for double ended chisels, types III and IV, shall be placed on the handle no less than 5 mm nor more than 12 mm from the junction of the handle and shank. The word "stainless" shall also be marked on instruments of foreign manufacture. As an alternate to the word "stainless", the use of a registered trade name identifying the steel as stainless shall be acceptable, provided prior approval is received from the cognizant contracting officer.

TABLE II. Instrument Number and Applicable Black's Formula

Type	Instrument Number	Black's Formula
I	81	20-9-6
I	83	10-6-6
I	84	20
I	85	15
I	86	10
II	41	--
II	42	--
III	3, 4	11½-15-3
III	5, 6	15-15-3
IV	40, 41	18-10-16

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hardness readings shall be made on test specimens representative

3.8 Workmanship. In addition to the requirements of GG-I-526, the operative ends shall be free of discolorations, pits or rounded edges. Workmanship shall be equal to or better than the standard sample (see 3.2 and 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the stainless steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier of the stainless steel, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Examination. The chisels shall be examined for defects including, but not limited to, those listed in table III. Examination for defects 102 and 103 shall be under four power (4X) magnification.

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TABLE III. Classification of Defects

Categories	Defects
Major A:	
101	Style and design not as specified for the applicable type and number chisel (see 3.5).
102	Cutting edge of operative end not sharp or has rounded edges.
103	Operative end not free from coarse grind marks, cracks, burrs, deep pits, feathers and other imperfections which affect serviceability.
104	Operative end finish not as specified.
Major B:	
151	Operative end dimensions not as specified.
Minor A:	
201	Handle or shank finish not as specified.
202	Handle or shank finish not free from coarse grind marks.
203	Handle not straight.
204	Identification marking not present, not properly located illegible, or incomplete.
205	Chisels not free from dirt, polishing compound, and foreign matter.
Minor B:	
251	Dimensions (other than 151) not within tolerances shown on drawing.
252	Handle serration dimensions not as specified.
253	Handle serrations missing, not uniform, or not complete.

The inspection levels and acceptable quality levels (AQLs) shall be listed in table IV. The unit of product for examination purposes shall be one chisel.

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TABLE IV. Examination Criteria

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Testing. Chisels shall be tested as specified in table V and 4.3.1. Inspection levels and acceptable quality levels (AQLs) shall be as listed in table V. The unit of product for test purposes shall be one chisel. Lots for hardness testing (heat treated operative ends) shall be as specified in 4.2.2.1.

TABLE V. Tests

Characteristic	Requirement	Test procedures	Inspection level	AQL
Chemical composition	3.3 and GG-I-526	GG-I-526	S-1*	**
Hardness	3.4 and GG-I-526	4.3.1	S-2	1.5
Copper sulfate test	GG-I-526	GG-I-526	S-2	1.0
Boil test	GG-I-526	GG-I-526	S-2	1

* A sufficient quantity of each lot of metal used in the construction of the chisels shall be selected for composition analysis in lieu of sample chisels, provided lot identity has been maintained and can be demonstrated to the Government.

** Acceptance number shall be zero for all sample sizes.

4.2.2.1 Lots for hardness criteria. For finished instrument (operative end) hardness testing, a lot or batch of instruments shall consist of instruments produced by one manufacturer at one plant, from the same material (same raw material heat and heat treatment) and from the same heat treatment lot or batch (instruments undergoing heat treatment at the same time and in the same heat treatment equipment.)

4.2.2.2 Sample unit usage. Where feasible, the same unit of product shall be used for the determination of two or more test characteristics.

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4.2.3 Inspection for preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table VI. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with AQL of 4.0 expressed in terms of percent defective.

TABLE VI. Examination of preparation for delivery

Examine	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

4.2.3.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

4.3 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.3.1 Hardness. Hardness shall be determined in accordance with ASTM E 18 (Rockwell) or E 92 (Vickers or Diamond Pyramid).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package. Each chisel shall be thoroughly cleaned and dried in accordance with the applicable method in MIL-P-116. Each chisel shall be wrapped in neutral tissue or other noncorrosive material and packaged in accordance with MIL-P-116, method IC-2. A preservative shall not be applied to the chisel.

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5.1.1.2 Intermediate package. Twelve unit packages containing chisels of one type and number only shall be packaged in a snug fitting box constructed in accordance with PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure of the box shall be adequate to prevent accidental opening.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Twenty four intermediate packages (288 units) containing chisels of one type and number only shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. Twenty four intermediate packages (288 units) containing chisels of one type and number only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. In addition to any marking specified in the contract or order, packages and shipping containers shall be marked in accordance with the requirements of Fed. Std. No. 123.

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5.3.2 Military agencies.

5.3.2.1 Interior packages. Each unit and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and number of chisel required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Approval of trade name if applicable (see 3.7).

6.2 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids or request for proposal (see 3.2, 3.6.1 and 3.8).

MILITARY CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6520-1402

Civil Agency Coordinating Activities:
VA - DMS
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